Operationalising Ambient Air Pollution Estimation

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The Problem

The current high-quality air pollution monitoring station network gin the UK is **spatially sparse with heterogeneous placement** and commonly suffers from **missing data temporally**. In England, only **103 Automatic Urban and Rural Network (AURN) monitoring stations** were online for NO_2 between 2014-2018, resulting in large areas of England missing localised measurements.

The Solution

We propose a **data-driven supervised machine learning model** to fill the missing air pollution concentration data temporally and spatially. We do this by training the model to **understand the relationship** between the environment around the monitoring station and the air pollution concentrations measured.

Model Framework

How Decision Trees Work

The type of machine learning model used is called a **decision tree.** Figure 1 shows a notional example decision tree for air pollution.

The first stage, **training**, is where the model analyses the **historical data** to **learn patterns**, in this case, the relationship between **environmental conditions** and **air pollution levels**. For example, the scenario of low air pollution concentrations at a time of high wind, shown as the green route in Figure 1. The data from existing monitoring stations is used as historical training data.

The model allows us to **predict air pollution levels** from the **known environmental conditions** based on the relationship learnt during training. This is known as the **prediction** stage.

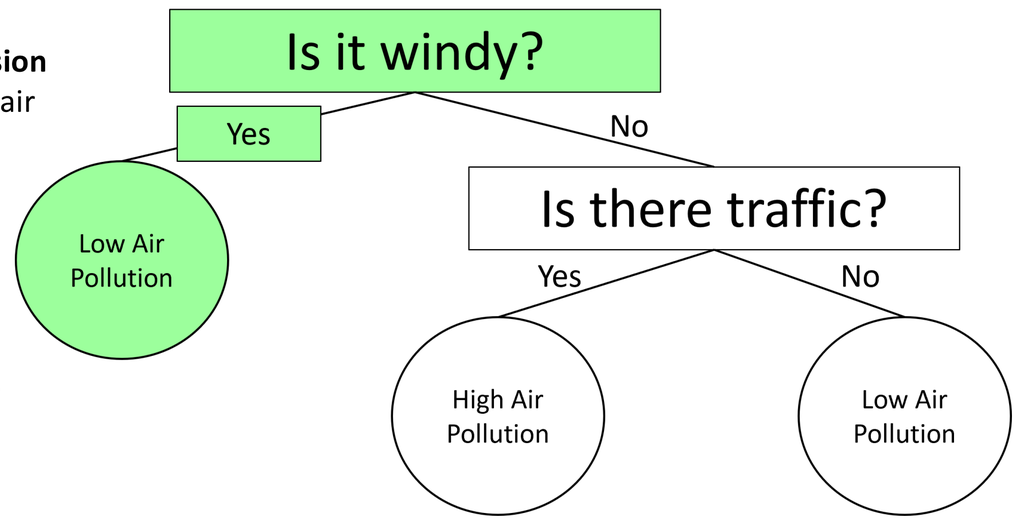

Figure 1: Example decision tree model based on traffic and wind. The decision tree model framework is based on the data in Table 1. Each route represents a possible air pollution scenario, given environmental conditions. The green route shows that when wind is high, air pollution is low.

Training Data

Wind	Traffic	Air Pollution
High	High	Low
Low	High	High
Low	Low	Low
High	Low	Low

Table 1: Example training data for the model detailing wind and traffic levels and the associated air pollution levels. Each table row represents a given scenario and corresponds to a route through the tree in Figure 1. The data can be extended to cover additional environmental conditions, such as those in Figure 3. The operational model discussed below uses 152 environmental conditions.

Understanding Our Past

Filling Missing Data Temporally

Since we have data describing the environmental conditions for each AURN station, we can **predict the air pollution concentrations at all timestamps**. Missing data, such as those in the monitoring station in the figure, can now be filled in, providing a **complete dataset** of **hourly** concentrations.

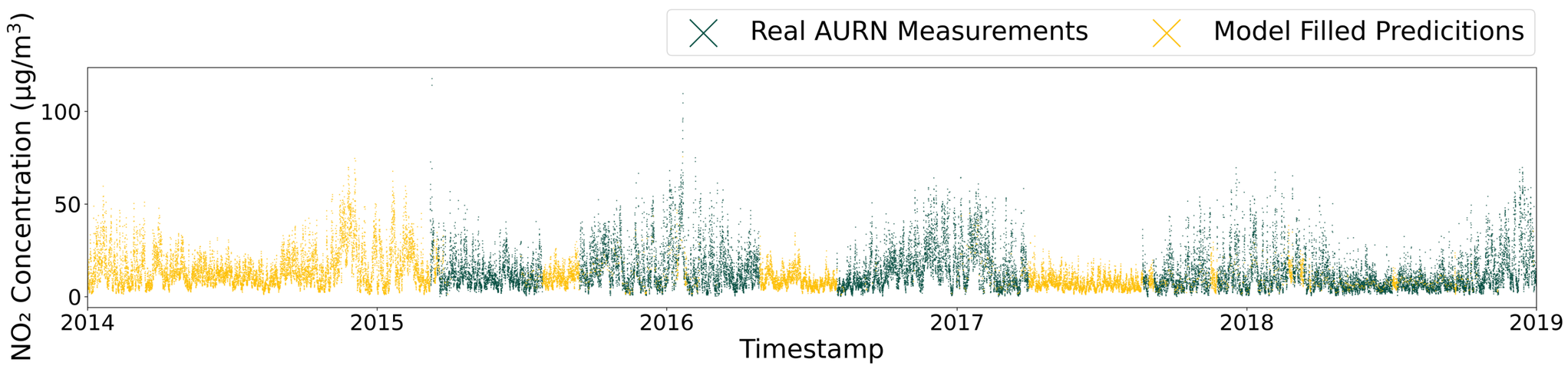

Figure 2: Chesterfield Loundsley Green monitoring station complete dataset, where model predictions (yellow) fill missing AURN measurements (green).

Filling Missing Data Spatially

Further to filling in data temporally, we can fill data spatially. Environmental data is available throughout England; therefore, we can **predict concentrations at all locations**. The model predictions equate to **355,827 stations at 1km intervals,** creating a dataset valued at **£70 Billion.**

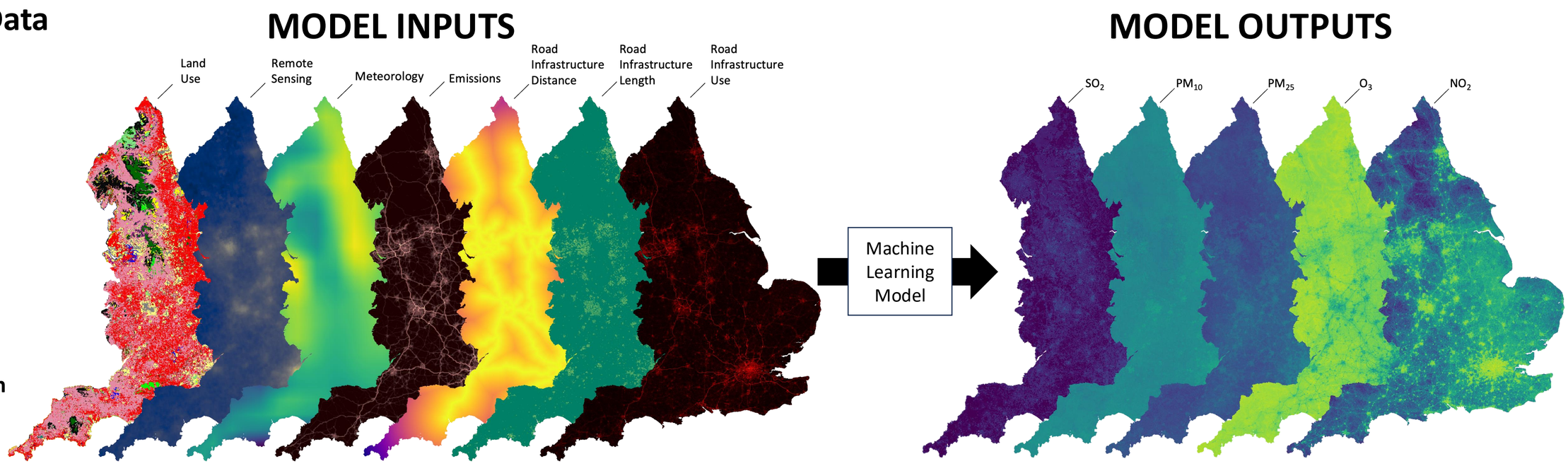

Figure 3: Example environmental conditions datasets (model inputs; left) and complete predicted air pollution concentration maps (model outputs; right).

Protecting Our Future

Exploring Hypothetical Situations

The model can also predict future air pollution concentrations for hypothetical situations. For example, the model can answer the question: **How would NO_2 air pollution concentrations change if a motorway was built in the East of England?** Any environmental condition used to train the model can be changed to explore hypothetical situations.

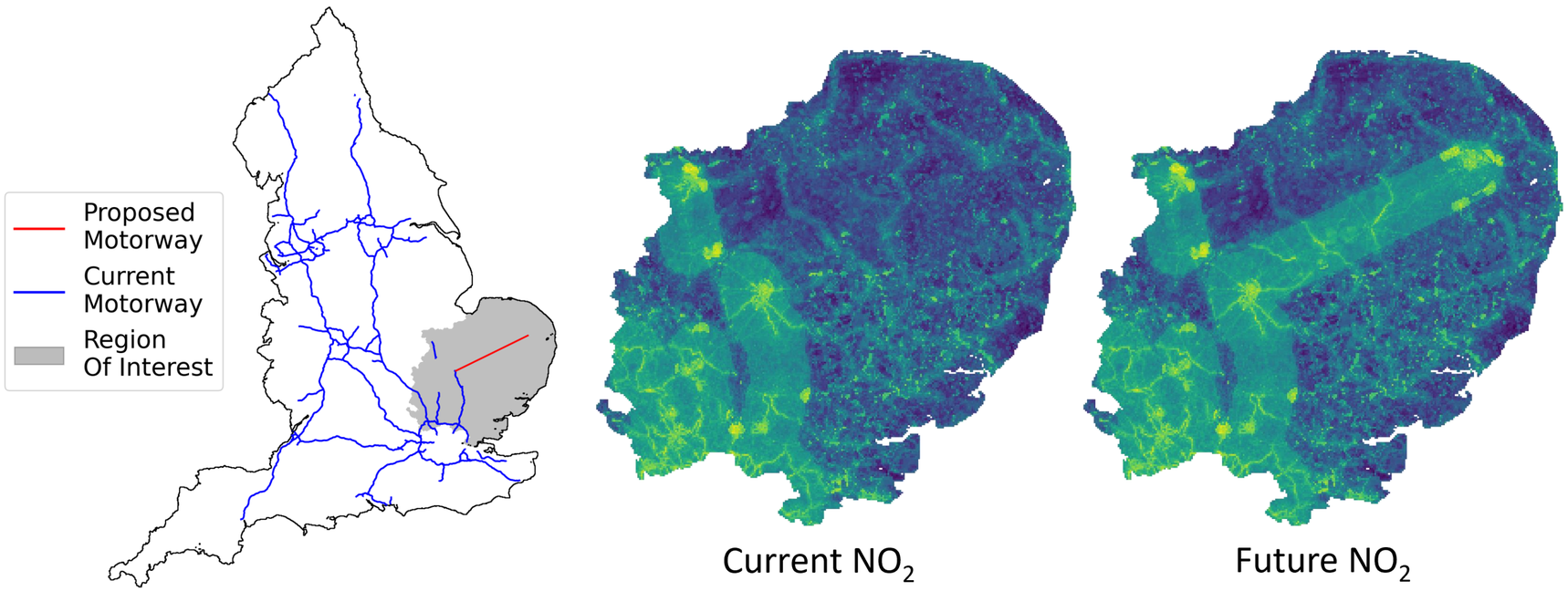

Figure 4: Current and proposed motorway locations across England (left). Predicted air pollution concentrations based on the current motorway network (middle). Predicted air pollution concentrations based on combined current and proposed motorway network (right).